

Double Integrals

ANSWER KEY



Iterated integrals over rectangles

- 1 Evaluate $\int_0^2 \int_0^3 (x^2 + y) dy dx$.
Inner: $\int_0^3 (x^2 + y) dy = 3x^2 + \frac{9}{2}$. Outer: $\int_0^2 (3x^2 + \frac{9}{2}) dx = [x^3 + \frac{9}{2}x]_0^2 = 8 + 9 = 17$.
 - 2 Evaluate $\int_1^2 \int_0^1 6xy^2 dx dy$.
Inner: $\int_0^1 6xy^2 dx = 3y^2$. Outer: $\int_1^2 3y^2 dy = [y^3]_1^2 = 8 - 1 = 7$.
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Double integrals over general regions

- 3 Evaluate $\int_0^1 \int_0^x (x + y) dy dx$, integrating over the triangular region $0 \leq y \leq x \leq 1$.
Inner: $\int_0^x (x + y) dy = x^2 + \frac{y^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2} dx = [\frac{x^3}{2}]_0^1 = \frac{1}{2}$.
- 4 Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \leq x \leq 1, 0 \leq y \leq 1$.
 $\int_0^1 \int_0^1 (4 - x - y) dy dx$: inner $= 4 - x - \frac{1}{2}$. Outer: $\int_0^1 (\frac{7}{2} - x) dx = \frac{7}{2} - \frac{1}{2} = 3$.